

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2012

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SHOOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

4. Which one of the following components of blood is involved in the clotting after injury?
- A. Plasma
 - B. White blood cells
 - C. Red blood cells

D. Platelets

The correct answer is D

1 [A] [B] [C] [D]

14. [A] [B] [C] [D]

24. [A] [B] [C] [D]

34. [A] [B] [C] [D]

11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of four boxes.

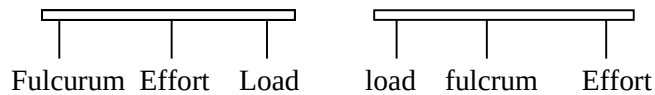
This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2012.

1. Which of the following is tuber crop?
 - A. Onion.
 - B. Carrot.
 - C. Pineapple.
 - D. Cabbage.
2. A beam balance is used for
 - A. Measuring volume of object.
 - B. Comparing mass of different objects.
 - C. Measuring height of objects.
 - D. Comparing size of different objects.
3. In which of the following groups of activities is friction **NOT** required?
 - A. Writing and erasing.
 - B. Skating and braking.
 - C. Flying and swimming.
 - D. Walking and riding.
4. Animals provides plants with?
 - A. Support and carbon dioxide.
 - B. Manure and oxygen.
 - C. Carbon dioxide and pollination.
 - D. Pollination and oxygen.
5. Which of the following statements about blood vessels in human body **CORRECTLY**?
 - A. All veins carry de-oxygenated blood.
 - B. Blood pressure in arteries is higher than veins.
 - C. Arteries have valves that control flow of blood.
 - D. Arteries have thinner walls than those of veins.
6. Which of the following **CORRECTLY** represents the path taken by food from the oesophagus?
 - A. Stomach-> small intestine-> large intestine->rectum->anus.
 - B. Small-> large intestine->stomach->rectum->anus.
 - C. Stomach->large intestine->small intestine -> rectum -> anus.
 - D. Large intestine->stomach->small intestine->small intestine ->rectum ->anus.
7. Which one of the following type of human teeth is **CORRECTLY** matched to its function?

Teeth	Function
A. Incisors	Tearing and cutting.
B. Canines	Tearing and crushing.
C. Premolars	Cutting and grinding.
D. Molars	Crusing.

8. A plant which has deep and widely spread root system is **LIKELY** to have
- Broad leaves.
 - Many leaves.
 - Succulent stems.
 - Weak stems.
9. Which of the following groups of crops consists of cereals **ONLY**?
- Peas, beans and sorghum.
 - Maize, rice and sorghum.
 - Millet, cow peas and sorghum.
 - Green grams, cow peas, maize.
10. The diagram below represent two levers **E** and **F** with different position of load, effort and fulcrum.



The levers represented by **E** and **F** are

- | E | F |
|----------------|--------------|
| A. Spade | crowbar. |
| B. Wheelbarrow | claw hammer. |
| C. Claw hammer | crowbar. |
| D. Spade | wheelbarrow. |
11. Which of the following a pair of plants is **CORRECTLY** matched **with** the type of roots system?

<u>Tap root</u>	<u>Fibrous root</u>
------------------------	----------------------------

- | | |
|------------|-------------|
| A. Grass | wheat. |
| B. Tomato | black jack. |
| C. Coconut | acacia. |
| D. Kale | onion. |
12. The following are statements about how different simple machines make work easier.
- Making the slop gentle.*
 - Reducing the effort applied.*
 - Changing direction of the force applied.*
 - Reducing friction.*

Which of the statements apply to a single fixed pulley?

- (i) and (iv).
 - (i) and (ii).
 - (ii) and (iii).
 - (iii) and (iv).
13. Which of the following groups of animals consists **ONLY** of invertebrates?
- Crabs, newt and snail.
 - Centipede, snail and spider.
 - Newt, centipede and slug.
 - Slug, frogs and crabs.

14. Which of the following statements is TRUE?

- A. Fish have constant body temperature.
- B. Birds have varying body temperature.
- C. Young ones of amphibians live on land.
- D. Reptiles breathe through the lungs.

15. Which one of the following are examples of herbivores, carnivorous and omnivores?

Herbivore	Carnivore	Omnivore
A. Hippopotamus	crocodile	chimpanzee.
B. Zebra	lion	vulture.
C. Gorilla	leopard	hyena.
D. Buffalo	baboon	rhino.

16. Which of the following pairs consists ONLY of animals with scales?

- A. Toad and duck.
- B. Turtle and salamanda.
- C. Turtle and duck.
- D. Salamanda and duck.

17. The following are some practices that help prevent the spread of water borne diseases.

- i) *Using toilets for urination.*
- ii) *Washing fruits before eating.*
- iii) *Washing hands before eating.*
- iv) *Wearing gumboots when walking in stagnant water.*

Which practices will help control the spread of bilharzia?

- A. (i) and (ii).
- B. (i) and (iv).
- C. (ii) and (iii).
- D. (iii) and (iv).

18. Transfer of heat by convection takes place in

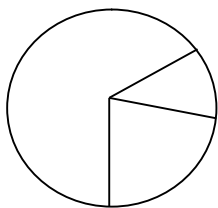
- A. Gases and liquid.
- B. Solids and liquid.
- C. Solids and vacuum.
- D. Gases and vacuum.

19. The diagram below represents the processes I, II, III and IV that are involved in change of states of matter.

Which of the following **CORRECTLY** represents the processes?

I	II	III	IV
A. Melting	freezing	evaporation	condensation.
B. Condensation	evaporation	melting	freezing.
C. Evaporation	melting	freezing	condensation.
D. Melting	freezing	condensation	evaporation.

20. The pie chart below shows proportions of G, H, and K that components of air



Which proportion **CORRECTION** represents a gas used in fire extinguishers and a gas is used in an electric tube?

Used in Fire extinguisher	used in electric tube
A. K	H
B. G	K
C. J	H
D. G	J

21. Which of the following factors make wax to float on water?

- A. Shape of materials.
- B. Type of material.
- C. Size of material.
- D. Mass of material.

22. In which of the following are both materials **CORRECTLY** grouped as magnetic and float and non-magnetic?

Magnetic	non-magnet
A. Copper	zinc.
B. Zinc	steel.
C. Aluminium	copper.
D. Steel	aluminium.

23. The following are activities carried out to investigate the effect of heat on a liquid

- i) *Placing the bottle on a source of heat.*
- ii) *Filling the bottle with coloured water.*
- iii) *Fixing a straw on the cork.*
- iv) *Closing the bottle tightly with the cork.*

Which one of the following is the **CORRECTLY ORDER** of the activities?

- A. (i) (iii) (iv) (i).
- B. (iii), (iv) (ii) (i).
- C. (ii) (i) (iii) (iv).
- D. (iii) (ii) (i) (iv).

24. The diagram below shows a set up that can be used to investigate the effect of heat on matter.

The effects investigate is

- A. Gases change state when heated.
- B. Solid expand when heated.
- C. Gases expand when heated.
- D. Solids change state when heated.

25. The diagram below represents a set-up in which a ruler is fixed to a desk and its free end pulled to position L, M, and N. At each position the free end is released to produce sound.

The set-up can used to investigate

- A. Special sounds.
- B. Loud and soft sound.
- C. Direction of sound.
- D. Noise pollution.

26. A card made in the shape of letter **E** and numbers 1, 2 and 3 written on it was placed in front of mirror as shown in the diagram below.

Which of the following **CORRECTLY** represents the image as seen in the mirror?

27. The reason why improved *jikos* have a clay lining is to

- A. Reduced the amount of heat loss.
- B. Reduced the space for charcoal.
- C. Make charcoal light faster.
- D. Make the jiko last longer.

28. Which of the following practices will NOT results in the person getting an electric shock?

- A. Touching sockets with wet hands.
- B. Putting pencils in sockets.
- C. Overloading the sockets.
- D. Putting iron nails in the sockets.

29. Which of the following groups of energy are renewable?

- A. Charcoal, kerosene, sawdust.
- B. Water, fire wood, animal dung.
- C. Diesel, wind, natural gas.
- D. Biogas, petrol, coal.

30. Which of the following pairs of food is preserved by smoking, salting and drying?

- A. Meat and mushrooms.
- B. Beans and fish.
- C. Mushrooms and beans.
- D. Meat and fish.

31. Which of following is the **MAIN** reason why eating bread after expiry date is discouraged? The bread could have

- A. Lost nutrients.
- B. A bad taste.
- C. Some micro-organisms.
- D. Changed its colour.

32. Which of the following pairs consists of disease caused by lack of minerals in the diet?

- A. Anaemia and rickets.
- B. Kwashiorkor and anaemia.
- C. Marasmus and kwashiorkor.
- D. Rickets and marasmus.

33. A hilly area with deep channels is **LIKELY** to be experiencing

- A. Sheet erosion.
- B. Gully erosion.
- C. Rill erosion.
- D. Splash erosion.

34. The diagram below shows an improvised rain gauge.

The collecting bottle should.

- A. Have the funnel sealed to its mouth.
- B. Be wide to collect enough water.
- C. Be placed above the ground.
- D. Be transparent for easy reading.

35. The diagram below represents a model of the solar system prepared by class six pupils.

The mistake made in the model was

- A. Using thread of wires.
- B. Placing a ball of clay next to the one representing the sun.
- C. Placing two balls of clay on one thread.
- D. Using balls of clay of different size.

36. The MAIN reason why mothers are encouraged breast feed their infants is because breast milk

- A. Boosts immunity of the infant.
- B. Is free from disease causing germs.
- C. Is at the right temperature for infants.
- D. Is easily digested by infants.

37. A person in the incubation period of HIV infection will show

- A. Signs and symptoms but test negative.
- B. No signs and no symptoms and test negative.
- C. No sign, no symptoms and test positive.
- D. Sign, symptoms and test positive.

38. Which of the following is a preventive measure against the spread of tuberculosis?

- A. Clearing bushes around houses.
- B. Boiling drinking water.
- C. Proper disposal of human waste.
- D. Living in a well-ventilated houses.

39. In which of the following are characteristic of cumulus and nimbus clouds is **CORRECTLY** placed

CUMULUS

NIMBUS

- | | |
|-------------------------|---------------------------------|
| A. Dark grey in colour | appears like bundles of cotton. |
| B. Mountainous in shape | white in colour. |
| C. White in colour | dark grey in colour. |
| D. Rain bearing | mountainous in shape. |

40. For two masses **X** and **Y**, the mass of **X** was twice that of **Y**. the diagram below shows a set up that was use to investigate how the two masses balance at position **Q, R,S** and **T** on the sea saw

Which of the following was a collection observation about the investigation?

Position of X	position of Y	observation made
A. Q	T	end P moved down.
B. R	T	end V moved up.
C. Q	S	seesaw balanced.
D. R	S	end V moved down.

41. The diagram below represents a simple electric circuit.

Which of the following energy transformation would take place when the switch is on?

- A. Chemical->electrical->heat -> light.
- B. Electrical-> chemical->flight ->heat.
- C. Chemical->electrical->flight->heat.
- D. Electrical-> flight ->chemical->heat.

42. Which of the following practices **LEAST** pollute the soil?

- A. Spraying crops to kill pests.
- B. Burning plant wastes in farm.
- C. Disposing of animal wastes in the farm.
- D. Burning plastics in the soil.

43. In an investigation, equal amount of soap was added to equal amount of water sample **W** and **Z** obtained from different source. The following observations were made:

- i) *W formed lather.*
- ii) *Z formed suspended particle.*

Which is of the following statements about **W** and **Z** is **NOT CORRECT**?

- A. W may have fewer minerals.
- B. Z may form lather after boiling.
- C. W can form scale in kettles.
- D. Z wastes soap when washing.

44. Which of the following consists of materials with a definite shape?

- A. Water, bottle, stone.
- B. Paper, pin, ink.
- C. Kerosene, cooking oil, milk.
- D. Sand, flour, sugar.

45. Which of the following practices will conserve soil?

- A. Using dry grass for mulching.

- B. Burning vegetation on farms.
 - C. Ploughing along the slope.
 - D. Burying plastic materials.
46. Soil becomes waterlogged due to
- A. Its colour.
 - B. The size of its particle.
 - C. The absence of small animals in it.
 - D. The amount of air in it.
47. Which of the following is CORRECT procedure that is used to obtain clear water from muddy water?
- A. Filtering and decanting.
 - B. Evaporation and filtering.
 - C. Decanting and filtering.
 - D. Evaporation and decanting.
48. Which of the following pairs of practices will control ticks, fleas and lice in livestock?
- A. Spraying and deworming.
 - B. Herding and dipping.
 - C. De-worming and rotational.
 - D. Dipping and spraying.
49. Which one of the following statements is TRUE about manure? They
- A. Increase the growth of weeds.
 - B. Release specific nutrients to the soil.
 - C. Destroy micro-organism in the soil.
 - D. Cause pollution of the soil.
50. The following are activities that pollute the environment
- i) Burning charcoal in the close room.
 - ii) Oil spillage from tankers.
 - iii) Dumping vegetable waste.
- Which activities pollute air?
- A. (i) and (iv).
 - B. (ii) and (iv).
 - C. (ii) and(iii).
 - D. (i) and (iii).